Latent Collaboration in Multi-Agent Systems

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Zou, Jiaru, et al. ”Latent Collaboration in Multi-Agent Systems” ICML2026 Spotlight

Background

Large Language Model-based Multi-Agent Systems (MAS)

- Extend single-model reasoning toward system-level intelligence
- Communication typically relies on *text* as the lingua franca (i.e., medium)

Limitations of Text-Based MAS

- **Efficiency:** Communication requires full decoding → **slow & token-heavy**
- **Expressiveness:** internal semantics compressed to vocabulary
- **Information Fidelity:** Re-encoding between agents causes **information loss**
- **Scalability:** System cost scales poorly as more agents/models participate

Motivation

Use LLMs' continuous latent space as a new form of "model language"

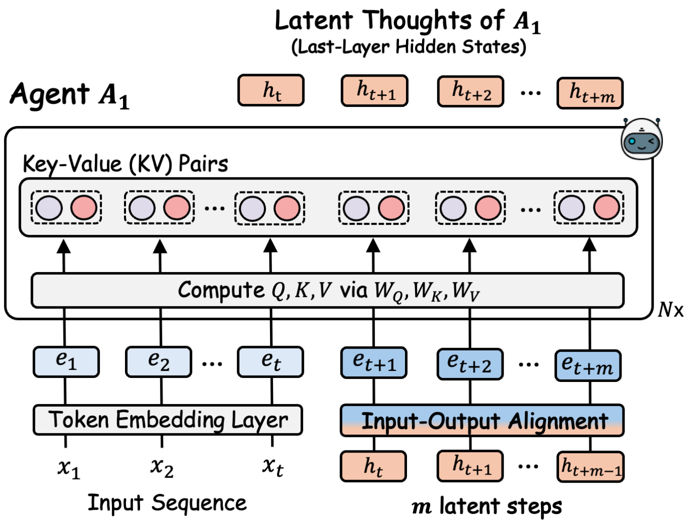
- **Reasoning Perspective:** Leveraging hidden representations within transformers to enable single model's internal latent chain-of-thought (CoT) reasoning.
- **Communication Perspective:** Employing KV caches or layer embeddings for information exchange across two models.

i.e., We aim to solve:

Can multi-agent systems achieve pure latent collaboration?

Building a Latent Collaborative Multi-Agent System (LatentMAS)

- **Auto-regressive Latent Thoughts Generation in Agents**



Given input $[x_1, x_2, \dots, x_t]$, inside each agent:

- Model generates directly **hidden states instead of tokens** for each reasoning step.
- Last-layer representation h_t becomes the next-step input — **no decoding needed**
- Produces a sequence of latent thoughts:

$$H = [h_{t+1}, h_{t+2}, \dots, h_{t+m}]$$

Agents "think" internally in latent space

Building a Latent Collaborative Multi-Agent System (LatentMAS)

- **Principle 1: Expressiveness Advantage**

Theorem 3.1 (Expressiveness of Latent Thoughts). *Under the Linear Representation Hypothesis on h (detailed in Assumption B.1), if the sequence of all latent thoughts with length m can be expressed losslessly through corresponding text-based reasoning, then the length of text (in tokens) needs to be at least $\Omega(d_h m / \log|\mathcal{V}|)$, where $|\mathcal{V}| > 1$ denotes the vocabulary size.*

Remark 3.2. Theorem 3.1 suggests that latent thoughts generation can be $O(d_h / \log|\mathcal{V}|)$ times more efficient than text-based reasoning. In addition, the expressiveness scales linearly with d_h, implying that larger models inherently exhibit greater latent reasoning capacity.

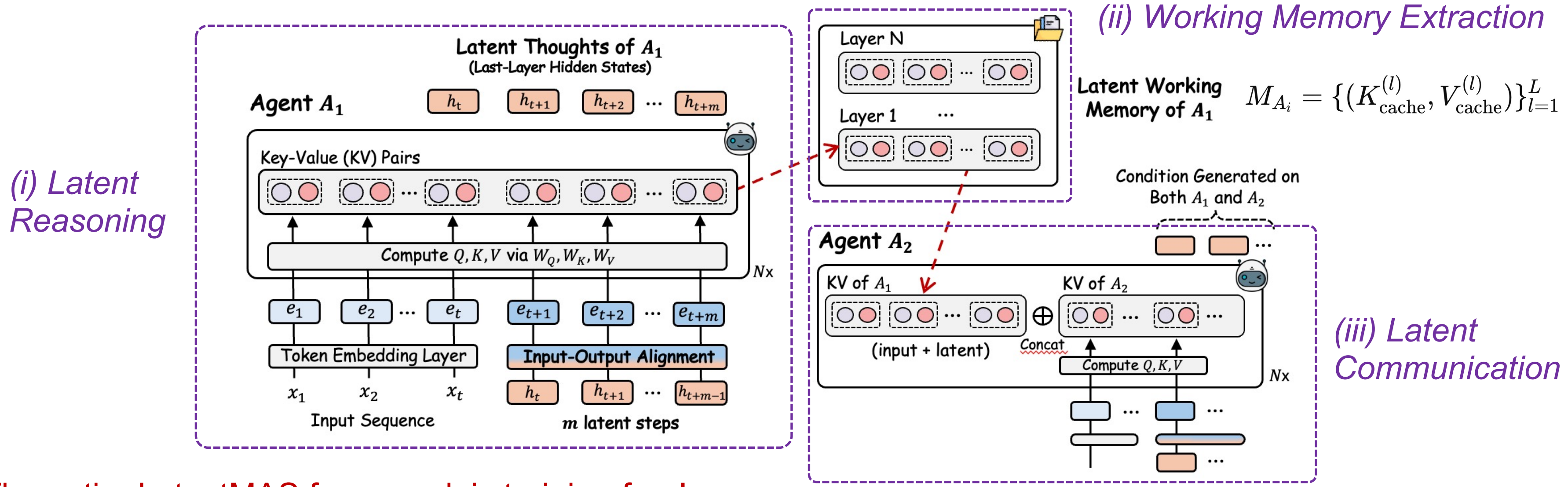
Latent thoughts are much more expressive than discreate texts (tokens) during model generation process!

*E.g., for Qwen3-4B / 8B / 14B models, latent thoughts generation can be **235.7 / 377.1 / 471.4 times more efficient** than text-based reasoning.*

Building a Latent Collaborative Multi-Agent System (LatentMAS)

- **Working Memory Preservation and Thoughts Transfer across Agents.**

After latent thoughts generation, LatentMAS transfers internal reasoning **without decoding**, preserving full semantics between agents.



The entire LatentMAS framework is training-free!

Building a Latent Collaborative Multi-Agent System (LatentMAS)

- **Principle 2: Lossless Information Transfer**

Theorem 3.3 (Information Preservation via Latent Working Memory). *In both latent and text-based reasoning, the outputs of an agent when receiving latent working memory from preceding agents are equivalent to those obtained when directly inputting the preceding agents' outputs.*

- **Overall Complexity**

Theorem 3.4 (LatentMAS Complexity). *The time complexity for each agent of LatentMAS is* $O\big((d_h^2 m + d_h m^2 + d_h t m)L\big)$, *where* t *is the input length of this agent, and* m *is the length of latent thoughts. In contrast, assuming Theorem 3.1, the time complexity for each agent of the vanilla text-based MAS needs to be* $\boxed{O\Big(\big(d_h^3 m \frac{1}{\log|\mathcal{V}|} + d_h^3 m^2 \frac{1}{\log^2|\mathcal{V}|} + d_h^2 t m \frac{1}{\log|\mathcal{V}|}\big)L + d_h^2 |\mathcal{V}| m \frac{1}{\log|\mathcal{V}|}\Big)}$ *to achieve the same expressiveness.*

Empirical Evaluation

Downstream Tasks: 6 general tasks, 3 reasoning-intensive tasks

Backbone Models: Qwen3-4B/8B/14B

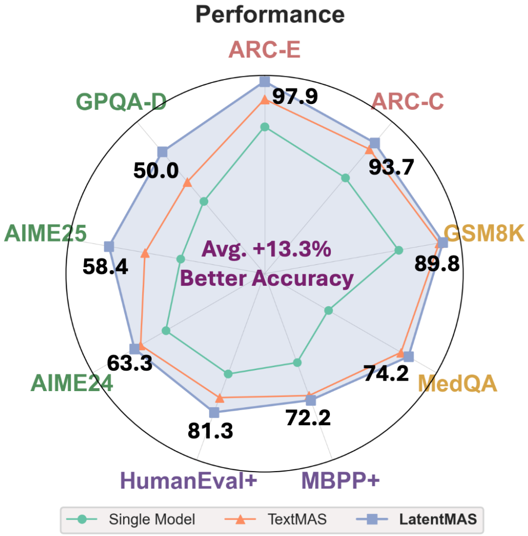
MAS Settings: Sequential & Hierarchical

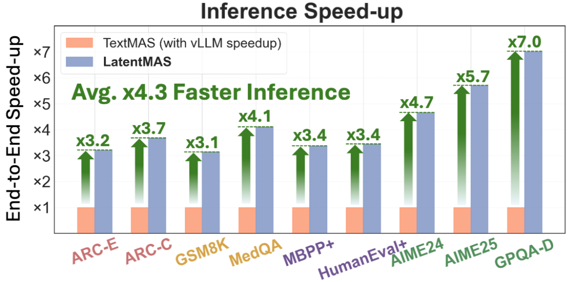
Overall Experimental Takeaways:

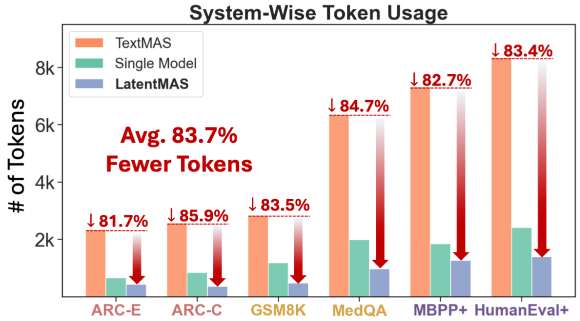
+13.3% accuracy vs TextMAS

-70.8% to -83.7% token usage

⚡ 4×-4.3× faster end-to-end inference







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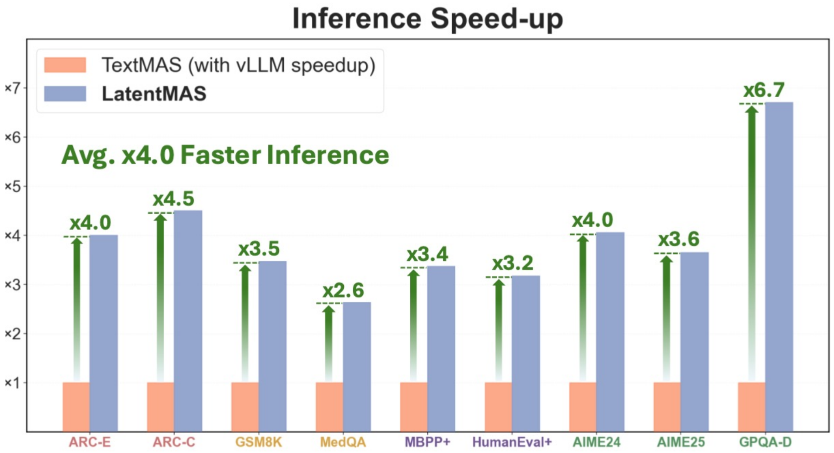
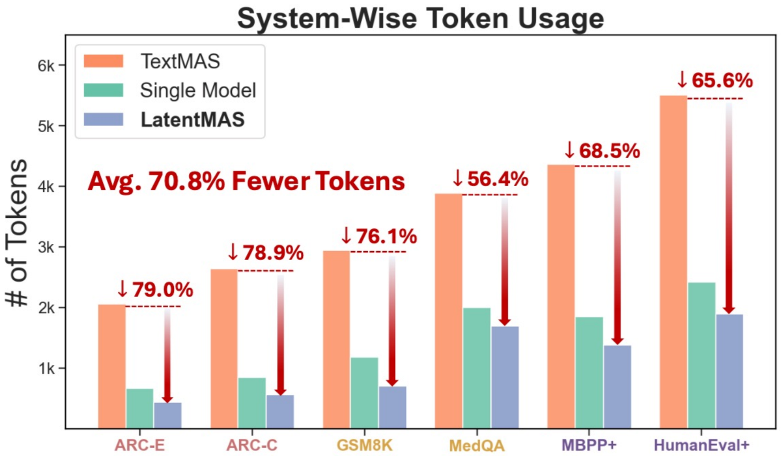
Additional Experiment

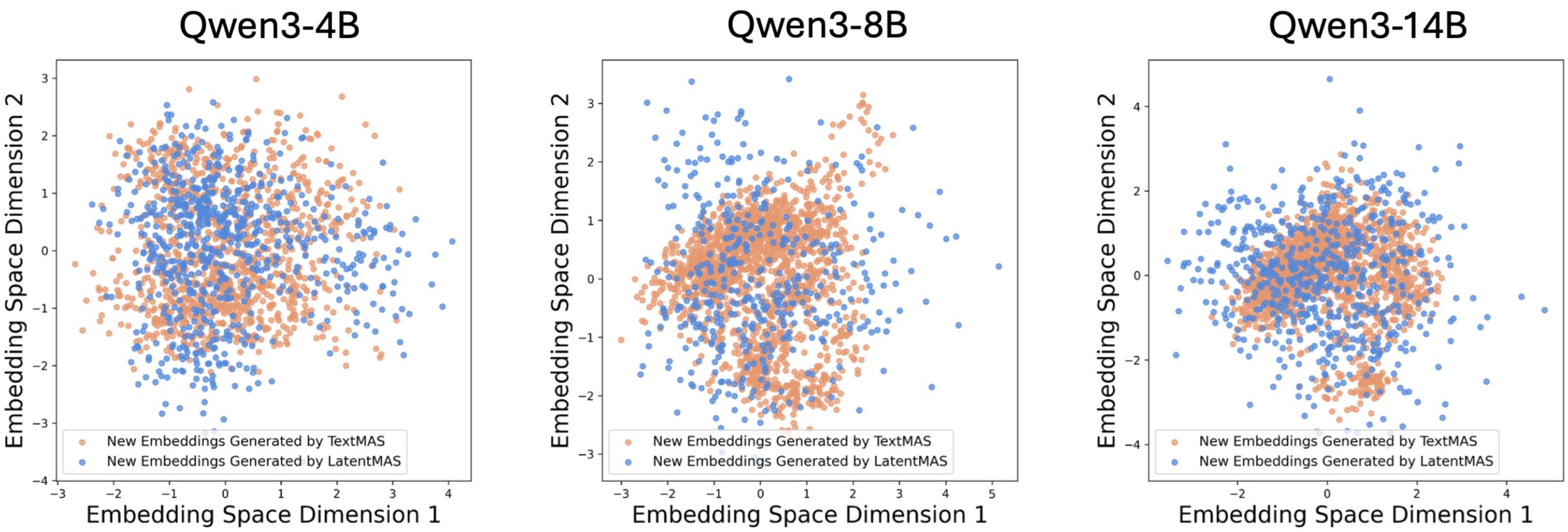
Efficiency gains of LatentMAS under the sequential MAS setting.





Additional Experiment

Illustration of the semantic meaning encoded by latent thoughts in LatentMAS



LatentMAS exhibits semantic consistency and greater expressive capacity than discrete text.

Takeaways

Paper: https://arxiv.org/pdf/2511.20639

Project: https://github.com/Gen-Verse/LatentMAS

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